

Community Data Work to Understand Environmental and Health Impacts of Data Centers

ERICKA CORRAL YANEZ[✉], DELANEY MARIE B. LINGAD, and YAJING LI, University of Denver, USA
ANGELIQUE RODRIGUEZ and BRANDON SCOTT, Globeville Elyria Swansea (GES) Coalition, USA
BENJAMIN XIE, University of Denver, USA

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1 Introduction: Data Centers as Sites of Environmental Injustices in the Age of AI

The recent wave of AI advancements have resulted in a massive proliferation of data centers. In the United States, data center development has become the largest infrastructure build out in peacetime history [2]. These data centers are often developed near communities disproportionately impacted by environmental harms without their awareness or participation [6, 36]. As consistent with previous environmental injustices in urban settings (e.g., sites of interstate highways, heavy industry, and pollution), the reason for this is cheaper land, no unique permitting process, easements, and perceived lack of community capacity and political will to facilitate responsible development [6, 36]. Data centers worsen air quality, become permanent urban heat islands [25, 33], and contribute noise pollution, further worsening environmental conditions in disproportionately impacted communities [39].

Data on the health and environmental impacts of data centers can help inform community members' efforts to organize and take actions to engage in the development process, mitigate impacts, and/or adapt to data center development. For example, data on how data centers' backup generators impact air quality, noise pollution, and temperature could help government agencies determine if a data center should be developed in close proximity to people or animals. However, data center developers and government agencies do not collect or make data on data center impacts readily available [12, 27].

Authors' Contact Information: Ericka Corral Yanez, ericka.corralyanez@du.edu; Delaney Marie B. Lingad, delaneymarie.lingad@du.edu; Yajing Li, yajing.li@du.edu, University of Denver, Denver, Colorado, USA; Angeliq Rodriguez, evolutionbluewave@proton.me; Brandon Scott, branolin@proton.me, Globeville Elyria Swansea (GES) Coalition, Denver, Colorado, USA; Benjamin Xie, benji.xie@du.edu, University of Denver, Denver, Colorado, USA.

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There is a need and opportunity for computer-supported cooperative work (CSCW) researchers to design approaches which build community advocates’ capacities to use data to understand the health and environmental impacts of data centers, support community organizing, and inform responsible policymaking. Following design justice [9] and participatory research approaches [20], this involves collaborations that meaningfully engage community advocates in the collection, analysis, interpretation, dissemination, and use of data such that data insights can augment existing efforts and practices. We frame data centers as environmental injustices in the AI age [6, 36], and make connections with prior community-engaged research in CSCW. We then provide a case study describing an ongoing collaboration between community advocates and HCI researchers to measure and disseminate environmental impacts of a forthcoming data center in and near the disproportionately impacted neighborhoods of Globeville, Elyria, Swansea (GES) in Denver, Colorado, USA. We conclude by identifying future research directions that use technology to support data work with disproportionately impacted communities which may become sites for data centers.

2 Background: Data Centers and Community-Engaged CSCW Research

2.1 Disparate Environmental Impacts of Data Centers

In this section, we summarize key environmental and public health impacts of data centers on nearby communities.

Energy Consumption: New and proposed data centers will use significantly more energy than existing ones, posing a risk to infrastructure, utility rates, and climate goals. AI data centers are typically *hyperscale* data centers with a capacity of 100+ MW [22], enough energy to power over 100,000 US households [21]. By 2028, data centers in the US are expected to consume 6.7-12% of all energy in the US [35], a 1.5-2.7x increase from five years prior. This increase in energy demand is straining electric grid at a time when the many regions in the world face greater heat waves and wildfires [1, 5].

Water Consumption: Most data centers require potable water for cooling purposes. Per industry standards, data centers must be running (their uptimes) for at least 99.671-99.995% of the time [8, 17]. Therefore, they cannot decrease water usage even during periods of drought or wildfires. Furthermore, data centers which access groundwater may decrease water quality and availability, especially for surrounding communities [13]. Data centers may contribute particular pressure to nearby water systems and environments [31].

Air Quality: Industry requirements for data centers’ uptimes constitutes the need for on-site energy generation capabilities, often with diesel generators during grid disruptions [26]. For example, the QTS Aurora-Denver 1 Data Center is looking to add 98 diesel backup generators [4]. These generators are tested regularly, typically resulting in regular air and noise pollution [4, 28].

Temperature Increases: A hyperscale data center is estimated to increase temperatures within one mile by up to 16°F (9°C) and temperatures within 3 miles by up to 6°F (7°C) [24]. For example, a 169 MW data center in Chandler, AZ increased downwind temperature by 4°F (2°C) which increases the need for air conditioning and therefore energy consumption and heat-related illnesses [34].

Noise Pollution: Data centers’ cooling systems and generators emit low-frequency noise which can harm people and animals [3, 30]. “Eerie hum[s]” from data centers are among the most common complaints from residents who live near data centers. A study done in Prince Williams County, VA, part of the world’s densest data center hub, found that data centers built within 1,000 ft could increase noise by 20 decibels (dB) [23]. In addition to audible sound, data centers also produce low-frequency noise which can be inaudible to people, yet still harmful, and can pass through solid objects (e.g., walls) with minimal energy loss [3, 15]. This ability to pass through walls with ease means that mitigation

of low-frequency noise must be done at the source and is therefore the responsibility of the data center developers and operators. Geographic features may further exacerbate noise issues with high winds and temperature inversions from the topography causing sound to travel further [14]. While low-frequency noise may not cause hearing loss, it is associated with headaches, sleep issues, and cognitive strain. Exposure to excessive noise can also impact animals' hormonal levels and behaviors.

2.2 Community-Engaged Research in CSCW

Participatory action research (PAR) is a collaborative research approach where researchers and community members work together to define a problem, develop a shared understanding, and take action toward change. PAR emphasizes shared learning and the addressing structural power dynamics and responsibilities with those most affected by the problem [20]. Jones et al. [20] describe participatory action research as an approach that combines ongoing learning with community involvement in deciding and enacting change, while also supporting participants in developing their own theories of change and ability to understand and interpret information. This is relevant to environmental justice work because it frames data collection as part of an ongoing process through which communities can identify concerns, interpret data, and decide how that data should support collective action.

Participatory design can reproduce exclusion even when inclusion is the goal, particularly when researchers fail to create shared spaces that empower community partners or establish common ground among stakeholders [11]. Community-collaborative research can also be understood as relational work, where relationships are not something that happens before the research, but part of the research itself [37]. This perspective carries obligations of accountability that can extend beyond completing a research project or grant [37]. Long-term participation allows researchers to develop a more complete understanding of stakeholders, practices, norms, and institutional arrangements while building trust that shorter engagements usually can't [20].

This research suggests that designing environmental sensing and data collection technologies with communities such as Globeville Elyria-Swansea (GES) should involve more than inviting residents to contribute data or provide feedback on a system that has already been defined. Instead, the work should create sustained infrastructure through which residents can help determine what should be measured, what constitutes evidence of harm, how environmental measurements should be interpreted alongside lived experiences, who controls and accesses the data, and what actions should follow from that data.

A common concern across this research is the difficulty of moving from community knowledge to institutional action. [16]'s "Voice to Vision" treats civic data infrastructure as a connection between community input, institutional decision-making, and public decision-making. Their study found that participants wanted to see themselves represented in collected data and also understand how their contributions affected decisions and outcomes. The paper also argues that visualizations are not neutral representations. They help shape what is recognized as legitimate evidence and can act as boundary between professional practices and community lived experience [16]. Participatory research around politically contentious topics also requires attention to trust, competing forms of practices, and potential harms. Researchers should build on existing relationships, allow time for trust to develop, remain flexible about how problems are defined, support mutual learning, and anticipate possible harms [40]. All in all, this research positions our work as a way to design a community-governed environmental data infrastructure that supports a pathway from environmental data collection and analysis with community experts to organize and develop policy intervention. This can help develop meaningful connections between environmental measurements, residents' lived experiences and health concerns, community priorities, and government or industry responses to advance mutual goals.

3 Case Study: Collaborating with Community Organizations and Policymakers in Elyria-Swansea

In this section, we summarize our ongoing research with Denver’s adjacent Globeville, Elyria, and Swansea (GES) neighborhoods. GES neighborhoods are a predominantly Hispanic and lower-income community with a history of environmental injustices and health disparities [29]. In 2016, the GES ZIP code was ranked the nation’s highest for overall environmental housing risks [38]. These results are reflective of the community’s proximity to an oil Refinery, the pet food plant, major highways, and railroad tracks, among many other industrial businesses [29]. The proximity to these facilities exposes locals to volatile organic compounds, nuisance odors, excessive noise, poor water quality, and contaminated soil [29, 32]. Alongside environmental conditions, those in the GES community disproportionately experience higher mortality rates, adult asthma, obesity, type II diabetes, gestational diabetes during pregnancy, congenital abnormalities, and disabilities [10]. Additionally, residents experience disparities in healthcare access, having higher rates of uninsured individuals and individuals behind on preventative healthcare [10].

A three building, 60 MW data center is currently under construction in the Elyria neighborhood, with the first 18 MW building set to be active within four months. This data center is located in a census block which ranks above the 94th percentile in the state for environmental burdens [7]. This data center is located near several gathering points for vulnerable populations including the elderly, adolescents, ill, and low-income community members: the data center is across the street from high-density affordable housing, an elder care facility, and a park; it is 500 feet (152 m) from a recreation center and community health clinic, and within 1900 feet (579 m) from a library. The community backlash from the development of this data center without their awareness resulted in a 12 month moratorium on data center development in the city [18].

Our ongoing research includes the development of low-cost sensors and dissemination of existing air quality data which residents can use to measure and mitigate environmental and health impacts of the data center. This involves measuring low-frequency noises emitted from data centers and analyzing an existing near-reference air quality monitor 500 feet (152 m) away. The low-frequency monitors use commercially available differential pressure sensors, Raspberry Pi computers, tubing, and other low-cost components to collect and store measurements over time¹. Because the monitors are relatively inexpensive, we plan to place several of these monitors at different parts near the data center property line. Community members will be involved in deciding where and when measurements are collected, especially near homes, clinics, parks, and other places where residents live or gather. Rather than asking residents only to host the monitors, we also plan to engage them in documenting when data center operations occur (e.g. testing or use of diesel generators), and compare their observations with the measurements collected by the sensors. Bringing these forms of information together may help residents identify patterns, discuss what should count as evidence of harm, and decide how the data could support public health initiatives, community organizing, and/or policy action. We plan to disseminate this information through a website accessed by mobile and a touchscreen at the lobby of the neighborhood health clinic in both Spanish and English. These findings can both advance scientific understandings of AI (data centers’) impacts on society and city and state elected officials on responsible policymaking on data center development. This research can advance our participatory action research practices related to data-intensive work with disproportionately impacted communities.

¹<https://github.com/starfishprime101/Infrasound-Monitor>

4 Conclusion: Discussions on Future Research

In this workshop, we hope to engage the CSCW community in identifying ways to research with community advocates to understand and meaningfully engage in the development process of data centers in disproportionately-impacted communities. Approaches could include participatory action research to develop community-centered approaches for collecting and disseminating environmental data around data centers as well as the design of informal learning spaces to build community capacity to organize against unregulated data center development. Key dimensions include concerns about data sovereignty, meaningful community collaboration, and well-being and safety of involved stakeholders (e.g. job opportunities, protection from retaliation). Despite trillions of dollars of investment in data centers, seven in ten people in the US oppose the development of data centers [19]. The CSCW community has an opportunity and perhaps prerogative to design with communities impacted first and worst by environmental injustices in the Generative AI age.

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